THE BASE-3 DIGIT OF 2^n AT A FIXED PLACE: LEAST PERIOD $2 \cdot 3^{k-1}$ AND EQUIDISTRIBUTION (CONJECTURE 1 OF OEIS A265210)

MATH0

Abstract. Let $T(n,k)$ be the base-3 digit of 2^n at the place 3^{k-1}; for fixed k, the sequence $n \mapsto T(n,k)$ over the rows n with $3^{k-1} \le 2^n$ is column k of the triangle OEIS A265210. We prove that column k has least period $2 \cdot 3^{k-1}$ for every $k \ge 1$, and that for $k \ge 2$ each of the digits $0, 1, 2$ occurs exactly $2 \cdot 3^{k-2}$ times in every block of $2 \cdot 3^{k-1}$ consecutive entries; this is Conjecture 1 of the entry (L. Edson Jeffery, 2015). The result follows quickly from the classical fact that 2 is a primitive root modulo 3^k, and its periodicity and equidistribution parts are immediate from results of de Faria and Tresser (2013) and of Aliyev (2023) on the last k ternary digits of 2^n, so we classify it as the first formal proof of a known result; the least period of a single column, which we did not find stated explicitly, is a short consequence of the same facts. Our proof avoids multiplicative orders and unit groups and uses three explicit congruences obtained by lifting the exponent. It is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the 84 listed terms.

1. Introduction

Write 2^n in base 3 and list its digits from the least significant one: row n of the irregular triangle A265210 of the OEIS [9] is this list (rows 0–5 are 1; 2; 1, 1; 2, 2; 1, 2, 1; 2, 1, 0, 1). Column k of the triangle consists of the k-th entries of the rows that have at least k entries, read from top to bottom; its entry in row n is the digit $T(n,k) = \lfloor 2^n / 3^{k-1} \rfloor \bmod 3$ of 2^n at the place 3^{k-1}. Column 1 is $1, 2, 1, 2, \ldots$, column 2 starts in row 2 with $1, 2, 2, 1, 0, 0, 1, 2, 2, 1, 0, 0, \ldots$, and column 3 starts in row 4 with $1, 0, 1, 2, 1, 2, 2, 2, 2, 1, 2, 1, 0, 1, 0, 0, 0, 0, 1, 0, \ldots$. The entry was submitted by L. Edson Jeffery on December 4, 2015, together with a conjecture that column k is periodic with period $2 \cdot 3^{k-1}$ and, for $k > 1$, equidistributed among the digits $0, 1, 2$ (Section 2). When we checked it in October 2026, the entry was still at revision #18 (December 6, 2015); it labels the statement a conjecture and contains no proof note. We prove it in the form of Theorem 1, with $2 \cdot 3^{k-1}$ the *least* period.

Prior work and status. The mathematics is classical, and we credit it as follows.

- It is well known that 2 is a primitive root modulo 3^k for every $k \ge 1$: the residues $2^n \bmod 3^k$ have least period $\varphi(3^k) = 2 \cdot 3^{k-1}$ and run through all units modulo 3^k (de Faria and Tresser cite [6, p. 81] for this). Lagarias uses this in the proof of his Theorem 1.4 [4, eq. (3.3)].
- De Faria and Tresser [2, §3.1, Lemma 3 and the paragraph following it] state that $2^n \bmod 3^k$ has minimum period $2 \cdot 3^{k-1}$ and that in any full minimum period each of the $2 \cdot 3^{k-1}$ ternary strings of length k not ending in 0 occurs exactly once as the string of the last k digits of 2^n. Since for $k \ge 2$ exactly $2 \cdot 3^{k-2}$ of these strings have a given digit at the place 3^{k-1}, the equidistribution and the period $2 \cdot 3^{k-1}$ of column k follow at once. Their preprint (July 2013) predates the conjecture.

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- Aliyev [1, Lemmas 3.1–3.3, the Note after Lemma 3.2, and Observation I] proves that 3^k exactly divides $2^{3^{k-1}}+1$, deduces that the order of 2 modulo 3^{n+1} is $2\cdot 3^n$, and shows that the last k ternary digits of 2^n are periodic and run through all endings that do not end in 0. His Lemma 3.1 is a weaker form of our Lemma 3.
- A comment of Jianing Song (2021) in A008776 [8] (the sequence $2\cdot 3^n$) records that $2\cdot 3^n$ is the order modulo 3^{n+1} of every integer congruent to 2 or 5 modulo 9.

The one part we did not find stated explicitly is that a single column (rather than the block of the last k digits) has *least* period exactly $2\cdot 3^{k-1}$. For $k\ge 2$ it follows in a few lines from the congruences $2^{3^{k-1}}\equiv -1$ and $2^{2\cdot 3^{k-2}}\equiv 1+3^{k-1}$ (mod 3^k) (Corollary 4), which is not a new idea. We therefore classify the result as the *first formal proof of a known result*. (One math0 referee classified it as settling the OEIS conjecture; the other, whose literature search produced the references above, judged it known; we follow the second.) All of Theorem 1 is proved in Lean, without multiplicative orders or unit groups; only Remark 13 is not formalized.

2. Statement

2.1. **The OEIS entry.** The name of A265210 [9] is `Irregular triangle read by rows in which row n lists the base 3 digits of 2^n in reverse order, n >= 0` (offset 0; 84 terms listed, rows 0 to 15). The COMMENTS section contains, verbatim,

> `Conjecture 1: The sequence in column k is periodic, with period p(k) = 2*3^(k-1) = A008776(k-1), k >= 1, and in which the numbers 0,1,2 appear with equal frequency, for each k>1.`

The comment is unsigned. The entry's author line reads `_L. Edson Jeffery_, Dec 04 2015`, and the revision history of the entry shows that he wrote the comment on that day (revisions #2–#5), so we attribute the conjecture to him, as the math0 statement file does.

2.2. **The theorem.** For $n\ge 0$ and $k\ge 1$ let $T(n,k)=\lfloor 2^n/3^{k-1}\rfloor \bmod 3$, let $C_k=\{n\ge 0 : 3^{k-1}\le 2^n\}$ be the set of rows having an entry in column k (an up-set of $\mathbb{N}$), and let $p(k)=2\cdot 3^{k-1}$.

Theorem 1. *Let $k\ge 1$.*

(i) *$T(n+p(k),k)=T(n,k)$ for every $n\in C_k$; and for every integer q with $0<q<p(k)$ there is $n\in C_k$ with $T(n+q,k)\ne T(n,k)$.*

(ii) *If $k\ge 2$, then for every $m\in C_k$ and every $d\in\{0,1,2\}$,*

$$\#\{\, n : m\le n<m+p(k),\ T(n,k)=d\,\}=2\cdot 3^{k-2}.$$

Theorem 1 is equivalent to the formal statement of Section 2. Part (i) says that $p(k)$ is a period of column k and that no q with $0<q<p(k)$ is one. In part (ii) the formal statement asserts only that the three counts are equal; since every $T(n,k)$ lies in $\{0,1,2\}$, the three counts add up to $p(k)=3\cdot 2\cdot 3^{k-2}$, so they are equal if and only if each of them is $2\cdot 3^{k-2}$. The Lean proof establishes the periodicity in (i) and the counts in (ii) for all $n,m\ge 0$, not only for rows of the column (for $n\notin C_k$ the value $T(n,k)$ is 0).

2.3. **Lean formalization.** The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A265210.lean`, reviewed for fidelity against the OEIS entry, and frozen (SHA-256 `56201c4b...fca62`) before any proof attempt. Its definitions, the statement, and the main data check read as follows (verbatim, except that the data list of `a_data` is abbreviated by "..."; the header, import, docstrings, blank lines and the other data checks are omitted):

```
namespace Math0.Statements.A265210
def row (n : ℕ) : List ℕ := Nat.digits 3 (2 ^ n)
def a (m : ℕ) : ℕ := ((List.range (m + 1)).flatMap row).getD m 0
theorem a_data : (List.range 84).map (fun i => a (i + 0)) =
```

```
    [1, 2, 1, 1, 2, 2, 1, 2, 1, 2, 1, 0, 1, 1, 0, 1, 2, 2, 0, 2, 1, 1, ..., 2, 2, 1, 1] := by
  decide +kernel
def T (n k : ℕ) : ℕ := (row n).getD (k - 1) 0
def IsColPeriod (k q : ℕ) : Prop :=
  ∀ n : ℕ, k ≤ (row n).length → T (n + q) k = T n k
def p (k : ℕ) : ℕ := 2 * 3 ^ (k - 1)
def statement : Prop :=
  ∀ k : ℕ, 1 ≤ k →
    (IsColPeriod k (p k) ∧ ∀ q : ℕ, 0 < q → q < p k → ¬ IsColPeriod k q) ∧
    (2 ≤ k → ∀ m : ℕ, k ≤ (row m).length → ∀ d e : ℕ, d < 3 → e < 3 →
      ((Finset.Ico m (m + p k)).filter (fun n => T n k = d)).card =
        ((Finset.Ico m (m + p k)).filter (fun n => T n k = e)).card)
end Math0.Statements.A265210
```

Encoding. `Nat.digits 3 x` lists the base-3 digits of x, least significant first, so `row n` is row n of the triangle. `a m` is entry m (offset 0) of the concatenation of rows $0, \dots, m$, which has more than m entries since every row is nonempty, so the default 0 of `getD` is never used. `T n k` is entry k of row n (with default 0), which is an actual entry of the triangle exactly when `k` $\le$ `(row n).length`. Row lengths are nondecreasing, so the rows with an entry in column k are $n_0, n_0+1, \dots$ for some $n_0 = n_0(k)$, and `IsColPeriod k q` says exactly that the column, read as the sequence $c(i) = T(n_0+i, k)$, satisfies $c(i+q) = c(i)$ for all $i \ge 0$. `p k` is $2 \cdot 3^{k-1}$, with no truncated subtraction since $k \ge 1$ wherever it is used.

Reading choices. (1) "Periodic, with period $p(k)$" is read as "$p(k)$ is the least positive period": every multiple of a period is a period, so naming one value refers to the least one. (2) "Appear with equal frequency" is read as equal counts of $0, 1, 2$ in every block of $p(k)$ consecutive entries; given periodicity, this is equivalent to equal counts in one period and to asymptotic frequencies $1/3$, so the truth value does not depend on this choice. Periodicity is asserted for $k \ge 1$ and equidistribution for $k \ge 2$, as in the conjecture (column 1 contains no 0). Both math0 referees judged the statement faithful.

Data checks. The definitions are anchored to the entry by lemmas proved with `decide +kernel`: `a_data` reproduces all 84 listed terms (rows 0–15), `row_example` the example rows 13–15, and `col1_data`, `col2_data`, `col3_data` the first terms of columns $1, 2, 3$ and the rows where columns $2, 3$ start; `sanity_small_columns` checks the periods 6 and 18 on the first 40 rows of columns $2, 3$ and the counts $(2, 2, 2)$, $(6, 6, 6)$ in their first periods. The proof does not use these checks.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A265210.lean`; Lean names are given in brackets. Throughout, for $j \ge 0$ we write

$$B = 3^{j+1}, \qquad M = 3^{j+2} = 3B, \qquad r = 2 \cdot 3^j, \qquad P = 2B = 3r = p(j+2),$$

and $d(n) = \lfloor 2^n/B \rfloor \bmod 3$, $s(n) = 2^n \bmod 3$. Column $j+2$ is the case $k = j+2 \ge 2$.

3.1. Digits and lengths.

Lemma 2 (`T_eq`, `len_iff`)**.** *For all $n \ge 0$ and $k \ge 1$, $T(n,k)$ as defined in Lean equals $\lfloor 2^n/3^{k-1} \rfloor \bmod 3$, and `k` $\le$ `(row n).length` holds if and only if $3^{k-1} \le 2^n$. In particular $T(n, j+2) = d(n)$.*

Proof. For $b \ge 2$, entry i (counting from 0, with default 0) of the digit list of x in base b is $\lfloor x/b^i \rfloor \bmod b$ (`Nat.getD_digits`); this includes the case where i is at least the length of the list, since then $x < b^i$ and both sides are 0. Also, the list has length at most i if and only if $x < b^i$ (`Nat.digits_length_le_iff`). Take $b = 3$, $x = 2^n$, $i = k-1$. □

3.2. **Lifting the exponent.**

Lemma 3 (`key`)**.** *For every $j \geq 0$ there is an integer $t \equiv 1 \pmod 3$ with $2^{3^j} = 3^{j+1}t - 1$.*

Proof. Induction on j. For $j = 0$ take $t = 1$. If $2^{3^j} = 3^{j+1}t - 1$, then

$$2^{3^{j+1}} = (3^{j+1}t - 1)^3 = 3^{j+2}\big(3^{2j+1}t^3 - 3^{j+1}t^2 + t\big) - 1,$$

and $t' = 3(3^{2j}t^3 - 3^j t^2) + t \equiv t \equiv 1 \pmod 3$. □

Corollary 4 (`E1`, `E2`, `E3`)**.** *For every $j \geq 0$:*

$$\textit{(E1)}\ \ 2^{2B} \equiv 1, \qquad \textit{(E2)}\ \ 2^B \equiv M - 1, \qquad \textit{(E3)}\ \ 2^r \equiv 1 + B \pmod M.$$

Proof. By Lemma 3 at $j + 1$, $2^B = Mt - 1$ for some integer t. Then $2^{2B} - 1 = M(Mt^2 - 2t)$ and $(M - 1) - 2^B = M(1 - t)$, which gives (E1) and (E2). By Lemma 3 at j, $2^{3^j} = Bt - 1$ with $t = 3s + 1$, and $2^r = (Bt - 1)^2$. Since $1 + 2t = 3(1 + 2s)$ and $B = 3 \cdot 3^j$,

$$(1 + B) - (Bt - 1)^2 = B\,(1 + 2t - Bt^2) = M\,(1 + 2s - 3^j t^2),$$

which gives (E3). □

3.3. **Consequences for the digits.**

Lemma 5 (`dig_congr`)**.** *If $a \equiv b \pmod M$ then $\lfloor a/B \rfloor \bmod 3 = \lfloor b/B \rfloor \bmod 3$.*

Proof. $\lfloor a/B \rfloor \bmod 3 = \lfloor (a \bmod 3B)/B \rfloor$ (`Nat.mod_mul_right_div_self`), and $3B = M$. □

Lemma 6 (`per`, `shift`, `two_pow_mod_three`, `mod_three_shift`)**.** *For every $n \geq 0$:*

(a) *$d(n + P) = d(n)$;*
(b) *$d(n + r) = \big(d(n) + s(n)\big) \bmod 3$;*
(c) *$s(n) \in \{1, 2\}$ and $s(n + r) = s(n)$.*

Proof. (a) $2^{n+P} = 2^n \cdot 2^{2B} \equiv 2^n \pmod M$ by (E1); apply Lemma 5. (b) By (E3), $2^{n+r} \equiv 2^n(1 + B) = 2^n + 2^n B \pmod M$, and $2^n B \equiv s(n)B \pmod{3B}$ because $2^n \equiv s(n) \pmod 3$. Hence $2^{n+r} \equiv 2^n + s(n)B \pmod M$, and by Lemma 5, $d(n + r) = \lfloor (2^n + s(n)B)/B \rfloor \bmod 3 = (\lfloor 2^n/B \rfloor + s(n)) \bmod 3 = (d(n) + s(n)) \bmod 3$. (c) $s(0) = 1$ and $s(n + 1) = 2s(n) \bmod 3$, so $s(n) \in \{1, 2\}$ by induction. Since $2^r = 4^{3^j} \equiv 1 \pmod 3$, $s(n + r) = s(n)$. □

Lemma 7 (`digit_flip`)**.** *If $2^n \equiv 1 \pmod M$, then $d(n) = 0$ and $d(n + B) = 2$.*

Proof. By Lemma 5, $d(n) = \lfloor 1/B \rfloor \bmod 3 = 0$, since $B \geq 3$. By the hypothesis and (E2), $2^{n+B} \equiv M - 1 = 3B - 1 \pmod M$, so $d(n + B) = \lfloor (3B - 1)/B \rfloor \bmod 3 = 2$. □

Lemma 8 (`proper_dvd`)**.** *If $g \mid 2 \cdot 3^{j+1}$ and $g < 2 \cdot 3^{j+1}$, then $g \mid 3^{j+1}$ or $g \mid 2 \cdot 3^j$.*

Proof. Write $g = y \cdot 3^b$ with $y \mid 2$ and $b \leq j+1$. If $y = 1$ then $g \mid 3^{j+1}$. If $y = 2$ then $2 \cdot 3^b < 2 \cdot 3^{j+1}$ forces $b \leq j$, so $g \mid 2 \cdot 3^j$. □

Lemma 9 (`count_three`)**.** *Let $f : \mathbb{N} \to \mathbb{N}$ and $\rho \geq 0$ satisfy $f(x) + f(x + \rho) + f(x + 2\rho) = 1$ for all x. Then $\sum_{x=m}^{m+3\rho-1} f(x) = \rho$ for every $m \geq 0$.*

Proof. Split $[m, m + 3\rho)$ into $[m, m + \rho)$, $[m + \rho, m + 2\rho)$ and $[m + 2\rho, m + 3\rho)$ and shift the last two to the first: the sum becomes $\sum_{x=m}^{m+\rho-1} \big(f(x) + f(x + \rho) + f(x + 2\rho)\big) = \rho$. □

Lemma 10 (`col_one`)**.** *Part (i) of Theorem 1 holds for $k = 1$ (and part (ii) is vacuous).*

Proof. Here $p(1) = 2$ and $T(n, 1) = 2^n \bmod 3$ by Lemma 2. Since $2^{n+2} = 4 \cdot 2^n \equiv 2^n \pmod 3$, 2 is a period. The only q with $0 < q < 2$ is $q = 1$, and row 0 lies in C_1 with $T(1, 1) = 2 \neq 1 = T(0, 1)$. □

3.4. **Columns $k \geq 2$.** Fix $j \geq 0$ and let $k = j + 2$, so that $C_k = \{n : B \leq 2^n\}$ by Lemma 2. Call $x \geq 0$ a *column period* [`Per`] if $d(n + x) = d(n)$ for all $n \in C_k$. By Lemma 2, q is a column period exactly when `IsColPeriod (j+2) q` holds.

Lemma 11 (`Per_mul`, `Per_gcd`)**.** *(a) If x is a column period, so is xt for every $t \geq 0$. (b) If q and Q are column periods and $g = \gcd(q, Q) < Q$, then g is a column period.*

Proof. (a) Induction on t. If $n \in C_k$ then $n + xt \in C_k$, since C_k is an up-set, so $d(n + x(t+1)) = d((n + xt) + x) = d(n + xt) = d(n)$. (b) Since $g < Q$, there is $c \geq 0$ with $qc \bmod Q = g$ (`Nat.exists_mul_mod_eq_gcd`), so $qc = Qu + g$ with $u = \lfloor qc/Q \rfloor$. For $n \in C_k$ we have $n + g \in C_k$, and by (a) applied to Q at $n + g$ and to q at n, $d(n + g) = d(n + g + Qu) = d(n + qc) = d(n)$. □

Proposition 12 (`col_ge_two`)**.** *For $k = j + 2$: (a) $T(n + p(k), k) = T(n, k)$ for all $n \geq 0$; (b) no q with $0 < q < p(k)$ is a column period; (c) for all $m \geq 0$ and $e \in \{0, 1, 2\}$, $\#\{n \in [m, m + p(k)) : T(n, k) = e\} = 2 \cdot 3^j$.*

Proof. Recall $T(n, k) = d(n)$ and $p(k) = P$. (a) is Lemma 6(a).

(b) Suppose $0 < q < P$ is a column period. By (a), P is a column period, and $g = \gcd(q, P) \leq q < P$, so g is a column period by Lemma 11(b). As $g \mid P = 2 \cdot 3^{j+1}$ and $g < P$, Lemma 8 gives $g \mid B$ or $g \mid r$, and by Lemma 11(a) B or r is a column period.

B is not a column period. Let $n = PB$. Then $n \in C_k$, since $B < 2^B \leq 2^{PB}$, and $2^n = (2^P)^B \equiv 1 \pmod M$ by (E1). Lemma 7 gives $d(n) = 0 \neq 2 = d(n + B)$.

r is not a column period. Let $n = B$, which lies in C_k since $B < 2^B$. By Lemma 6(b),(c), $d(B + r) = (d(B) + s(B)) \bmod 3$ with $s(B) \in \{1, 2\}$, so $d(B + r) \neq d(B)$. This proves (b).

(c) Fix $e < 3$ and let $f(x) = 1$ if $d(x) = e$ and $f(x) = 0$ otherwise. For $x \geq 0$ put $a = d(x)$ and $s = s(x) \in \{1, 2\}$. By Lemma 6(b) at x and at $x + r$, and (c),

$$d(x + r) = (a + s) \bmod 3, \qquad d(x + 2r) = \big((a + s) \bmod 3 + s\big) \bmod 3 = (a + 2s) \bmod 3.$$

Since $s \not\equiv 0 \pmod 3$, the residues $a, a + s, a + 2s$ are distinct modulo 3, so exactly one of $d(x), d(x + r), d(x + 2r)$ equals e, i.e. $f(x) + f(x + r) + f(x + 2r) = 1$. Since $P = 3r$, Lemma 9 with $\rho = r$ gives the count $r = 2 \cdot 3^j$. □

Proof of Theorem 1. [`result`] For $k = 1$ use Lemma 10. For $k = j + 2 \geq 2$, Proposition 12(a),(b) is part (i), and (c) is part (ii), since $2 \cdot 3^j = 2 \cdot 3^{k-2}$; in particular the three counts are equal, as the formal statement asserts. □

Remark 13 (not formalized)**.** Theorem 1 implies that 2 is a primitive root modulo 3^k. Indeed, the order o of 2 modulo 3^k is a period of column k, since $T(n, k)$ depends only on $2^n \bmod 3^k$. If $g = \gcd(o, p(k))$ were less than $p(k)$, then g would be a period of column k by the argument of Lemma 11(b) (which works for every $k \geq 1$), contradicting Theorem 1(i) since $0 < g < p(k)$. Hence $p(k) \mid o$, while $o \mid \varphi(3^k) = p(k)$ by Euler's theorem. The classical route goes the other way: since 2 is a primitive root, $2^n \bmod 3^k$ runs once through all units modulo 3^k in each period [2, 4], and one counts the units by their digit at the place 3^{k-1}.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A265210.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A265210.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A265210.result`, of type `Math0.Statements.A265210.statement`; the auxiliary declarations, all private, are the ones named in brackets in Section 3 (plus `T_eq'`, the case $k = j + 2$ of `T_eq`, and the definition `dig` of d). The

final theorem and the audit command read as follows (verbatim, except that the docstring and a blank line are omitted):

```
theorem result : statement := by
  intro k hk
  rcases Nat.lt_or_ge k 2 with hk2 | hk2
  · obtain rfl : k = 1 := by omega
    exact ⟨col_one, fun h => absurd h (by norm_num)⟩
  · obtain ⟨j, rfl⟩ : ∃ j, k = j + 2 := ⟨k - 2, by omega⟩
    obtain ⟨h1, h2⟩ := col_ge_two j
    refine ⟨h1, fun _ m _ d e hd he => ?_⟩
    rw [h2 m d hd, h2 m e he]
end Math0.Frontier.A265210
#math0_settles Math0.Frontier.A265210.result Math0.Statements.A265210.statement
```

Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`.

Audit. The command `#math0_settles` (in `lean/Math0/Audit.lean`, for statements that math0 formalizes itself) fails unless (1) `statement` is a universe-monomorphic `Prop`-valued definition; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, `Lean.ofReduceBool` (from `native_decide`) and custom axioms; and (3) the type of `result` is syntactically `statement` or ¬ `statement`, up to metadata, with no unfolding. It reports

```
math0 audit: 'Math0.Statements.A265210.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A265210.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A265210.result' proves 'Math0.Statements.A265210.statement'
```

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [7] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [3] at commit `c77fdd4`, which this result does not use (the statement is math0's own formalization of the OEIS conjecture). To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, runs `lake build` on all math0 results (re-running every audit), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file is checked by `lake env lean Math0/Frontier/OEIS/A265210.lean` in `lean` (about 17 s on our machine).

References

[1] Y. N. Aliyev, *Digits of powers of 2 in ternary numeral system*, Notes Number Theory Discrete Math. **29** (2023), no. 3, 474–485, https://doi.org/10.7546/nntdm.2023.29.3.474-485.

[2] E. de Faria and C. Tresser, *On Sloane's persistence problem*, Exp. Math. **23** (2014), no. 4, 363–382, https://doi.org/10.1080/10586458.2014.910849; arXiv:1307.1188 (July 2013), whose numbering we use.

[3] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: https://github.com/google-deepmind/formal-conjectures.

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